Turgor soybean stress index: validation report

Drought and heat in soybean in Rio Grande do Sul and Paraná, Brazil, from public satellite and weather data. October 2026.

Summary

- **What it is:** a drought and heat stress index for soybean, weighted by crop stage, computed only from public satellite (MODIS, Sentinel-2) and weather (NASA POWER) data, with no farm sensors.
- **Backtest (2001-2023, 17,342 municipality-seasons):** it separates loss seasons (yield >10% below trend) from normal ones with AUC 0.86, against 0.78 for a rainfall index and 0.57 for ZARC. It misses 33% of real losses, against 44% for the rainfall index.
- **New seasons (2024 and 2025, never used to build the index):** AUC 0.67 against 0.56; losses missed 36% against 50%.
- **Lead time:** by 15 January the index already has the skill the rainfall index reaches only at season end, 8 to 12 weeks before harvest. On 15 February 2025, municipalities flagged as severe stress ended 30% below trend.
- **Limits, openly:** in Rio Grande do Sul the gain over the rainfall index is not proven at municipality level; loss probabilities were too confident in 2024-2025, mostly in Paraná; field-level validation still needs a co-op's data.

1. Method

For each municipality and season, the index adds, separately for the vegetative and the reproductive stage, three components standardised against the municipality's own history: water deficit (crop water demand not met by rain and soil water), degree-days above 30 °C, and the heat × drought interaction. It subtracts the NDVI anomaly (canopy vigour). The reproductive stage (flowering, pod set, seed fill) weighs 0.75 and the vegetative stage 0.25.

No weight was fitted to yield data. The weights come from soybean physiology. A separate test showed they match the FAO yield-response factors (0.82 reproductive) and the weight the data themselves would choose (0.75 in PR, 0.80 in RS); fitting the weight does not improve the result.

Real-time version: the index uses each municipality's typical stage windows (median of its other seasons) and real data latency (weather 3 days late, MODIS 16-day composites 23 days late). The numbers in this report are those the index would have given at the time.

Ground truth: IBGE municipal soybean yields (Produção Agrícola Municipal), detrended per municipality. Loss = yield more than 10% below trend. **Basis risk** = share of real losses a trigger misses when it pays as many municipality-seasons as there were losses.

2. Backtest, 2001-2023

Index	AUC (95% CI)	Losses missed	State-year correlation
Turgor index	0.86 (0.81-0.90)	33% (25-46%)	0.86 (0.78-0.92)
Rainfall index (Jan-Feb)	0.78 (0.70-0.84)	44% (33-60%)	0.68 (0.52-0.78)
NDVI only	0.77 (0.72-0.81)	44% (35-59%)	0.85 (0.76-0.91)
ZARC (official zoning)	0.57 (0.54-0.61)	64% (55-75%)	0.41 (0.09-0.65)

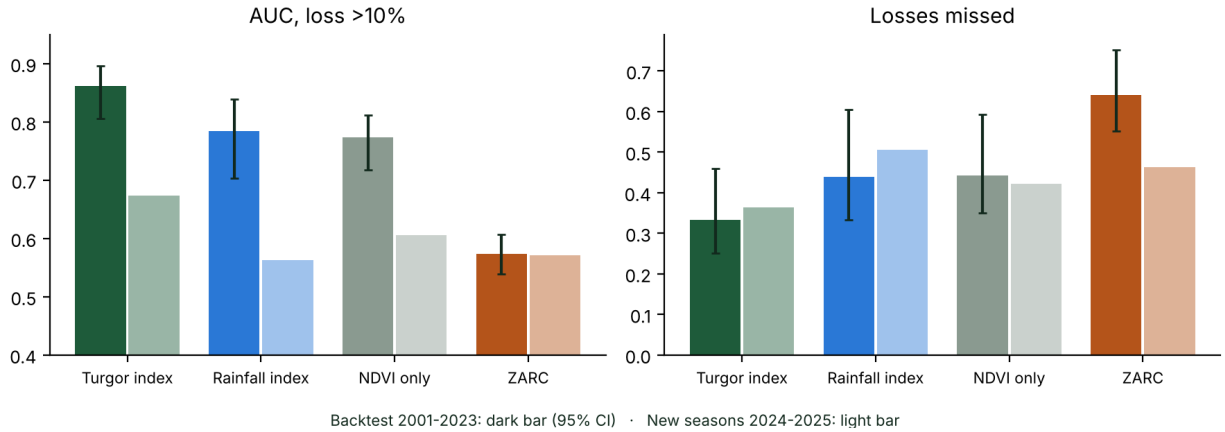
95% intervals from 1,000 resamples of whole harvest years: municipalities in the same season share weather and are not independent.

	RS: AUC	RS: Losses missed	PR: AUC	PR: Losses missed
Turgor index	0.90	26%	0.82	39%
Rainfall index (Jan-Feb)	0.86	31%	0.70	60%
NDVI only	0.83	36%	0.70	55%
ZARC (official zoning)	0.58	60%	0.54	73%

- **RS + PR and Paraná: clear gain over the rainfall index** (AUC +0.08, CI +0.03 to +0.13; in PR losses missed 39% against 60%).
- **Rio Grande do Sul: likely but not proven** at municipality level (AUC 0.90 against 0.86; CI of the difference -0.015 to +0.094). RS droughts are mostly rainfall droughts. We tested 250 m resolution, a soybean-only mask (MapBiomas) and vapour-pressure deficit: none changed the result. With 23 seasons, basis-risk gains below about 0.10 cannot be proven per municipality.
- **ZARC** barely separates loss seasons (AUC 0.57): it is a fixed, pre-planting zoning. Turgor adds in-season information to what the market already uses; it does not replace ZARC.

3. New seasons: 2024 and 2025

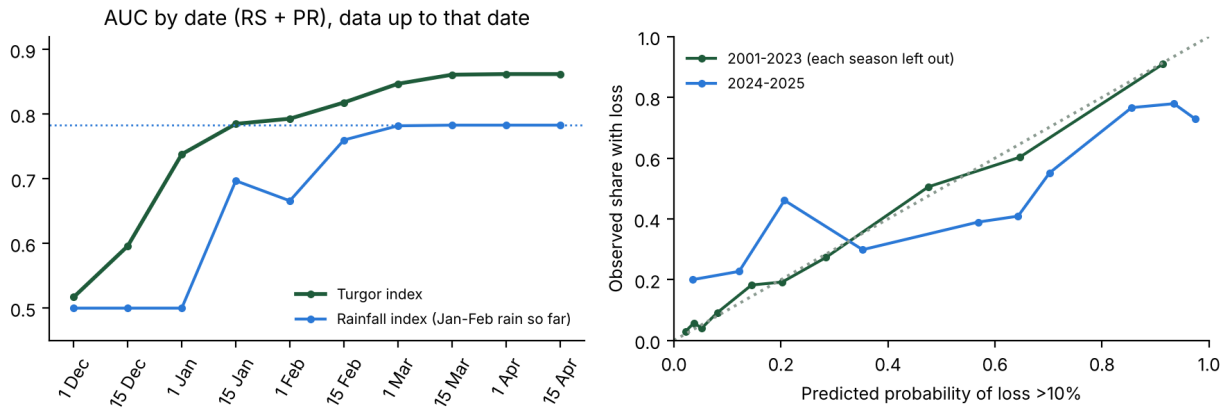
IBGE yields downloaded after the index was finished; trend and standardisation use 2001-2023 only. 1,542 municipality-seasons.

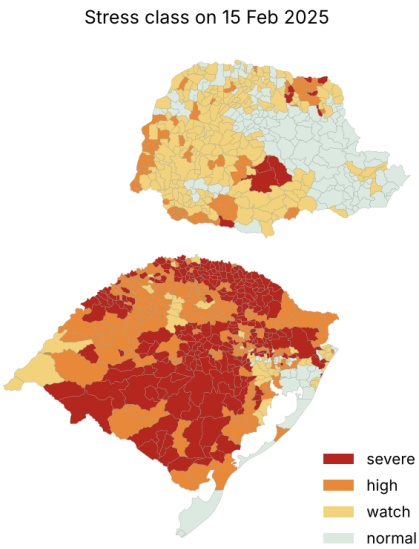


- The index misses 36% of real losses, against 50% for the rainfall index, 42% for NDVI and 46% for ZARC.
- Skill is lower than in the backtest (AUC 0.67 against 0.86). Two seasons is a small sample, and part of the 2024 RS loss came from the May floods at harvest, which no drought index should flag.

4. How early

The index rebuilt as it would have looked on each date, using only the data available that day.





Replay of 15 February 2025 (RS drought), with 2025 left out of the reference:

Class on 15 Feb 2025	Municipalities	Yield vs trend	With loss >10%
severe	227	-30%	83%
high	212	-24%	68%
watch	215	-5%	33%
normal	117	+1%	19%

AUC 0.81, about two months before harvest. From December 2026 the map is published every week at github.com/Fortirer/soy-stress-index/tree/main/monitor, dated, before harvest.

5. From index to money

Monotone (isotonic) maps per state turn the index into a probability of loss, an expected yield and an expected loss in R$/ha. On 2001-2023, predicting each season without using it, the probabilities are well calibrated (Brier score 0.126 against 0.204 for the historical rate: 38% better). On 2024-2025 the gain drops to 16% (RS 40%, PR -17%): the probabilities were too confident at both ends. They are therefore capped at 2-98% and presented as historical base rates, not guarantees.

6. Stage weights: physiology, literature and data agree

The reproductive weight of 0.75 was compared with the FAO yield-response factors for soybean (Doorenbos and Kassam, 1979: vegetative 0.2, flowering 0.8, yield formation 1.0, i.e. 0.82 reproductive), equal weights, reproductive only, and the weight chosen by the data season by season. The data choose between 0.65 and 0.90 (median 0.75 in PR, 0.80 in RS). Fitting the weight makes the result slightly worse (AUC 0.857 against 0.862). Three independent sources agree, and a weight fixed before looking at yields generalises as well as any fitted weight.

7. Next step: field-level validation

The field version uses each field's Sentinel-2 curve (10-20 m) and the weather at the field, with the same index definition. In a sample of 15 fields in RS, the drop in reproductive NDVI from 2021 to 2022 ranked municipalities like the IBGE loss and showed large differences between neighbouring fields (in Cruz Alta, 12% in one field and 51-61% in its neighbours) that a municipal index cannot see. Validation against real yields needs a partner co-op; reading the files, the test and the report are already in place.

8. Limits

- Rio Grande do Sul: gain over the rainfall index per municipality not proven (section 2).
- Loss probabilities too confident in unusual seasons; in PR worse than the historical rate in 2024-2025.
- Municipal scale: a municipality mixes many farms. The field version (Sentinel-2) is ready but can only be validated with a co-op's real yields.
- Drought and heat only. Excess rain, floods, hail, frost and disease are not in the index.
- Yield trend extrapolated to new seasons; NASA POWER rainfall on a ~50 km grid.
- 23 seasons of history: confidence intervals are wide and will narrow with each monitored season.

9. Reproducibility and review

Open code, data and results (MIT licence): github.com/Fortirer/soy-stress-index. Every number in this report is read from the files in *results/*, and the report is rebuilt by *scripts/23_public_report.py*. Critical reviews are welcome, especially from crop modellers and agricultural insurance actuaries: open an issue in the repository.

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